



CALIFORNIA BULRUSH

Schoenoplectus californicus (C.A. Mey.) Palla

Plant Symbol = SCCA11

Alternative Names

Common Names: bulrush, tule, black root, bullwhip, and giant rush

Scientific Names: *Scirpus californicus* (C.A. Mey.) Steud, *Scirpus californicus* var. *tereticulmis*

Description

General: Sedge Family (Cyperaceae). California bulrush, a native, herbaceous, rhizomatous perennial plant which forms dense colonies on mudflats, in open water and along shore water interfaces, is distinguished by long triangular or subterete stems from 5 to 10 feet tall. The leaves are slender V-shaped blades that are sheathed around the long stem. The stems are slightly angled with drooping branched terminal inflorescences. The flowers are arranged in a spikelet and resemble orange-brown scales. The inflorescence of California bulrush is a larger and more open panicle than hard stem bulrush (*Schoenoplectus acutus*), with 20 or more spikelets, and it is consistently a reddish-brown color. Bulrushes have clonal growth, with stout rootstocks and long, thick, brown rhizomes (underground stems) (USDA NRCS, 2003).

Distribution: California bulrush has a wide distribution across the southern United States and west coast. States where California bulrush is native include Illinois, Tennessee, South Carolina, Florida, Georgia, Alabama, Mississippi, Arkansas, Louisiana, Texas, Oklahoma, Kansas, New Mexico, Arizona, California, and Oregon. For current distribution, please consult the Plant Profile page for this species on the PLANTS Web site (USDA NRCS, 2003).

Adaptation

Bulrushes are often dominant emergent vegetation found in marshes and wetlands throughout temperate North America. California bulrush, which often hybridizes with hard stem bulrush, is abundant in freshwater marshes along the California coast from San Diego County to Napa County, inland in the Central Valley, occasional in the Mojave Desert, Texas to South Carolina, Florida, temperate South America and the Hawaiian Islands (Hickman, 1993). Although California bulrush can tolerate some salinity, it is mostly found in fresh or brackish water and thrives in hardiness zones 3a to 9b (Tiner, 1984). California bulrush is adapted to fine and medium texture soils with high anaerobic tolerance (USDA NRCS, 2003).

Uses

Conservation: California bulrush is used effectively to control shoreline erosion along streams, bayous, levee banks, canals and other places where land and water meet. Additionally, this species will form a vegetative barrier across open water reducing wave energy impacting shorelines. California bulrush is used extensively in marsh restoration, as well as streambank stabilization, wetland restoration, wildlife habitat establishment, as an edible crop (young shoots, pollen, seeds, rhizomes), and for matting. These native plants are especially good for stabilizing or restoring disturbed or degraded (including logged or burned) areas, for erosion and slope control, and for wildlife food and cover. Bulrushes may be less suitable for general garden use (Dahl & Johnson, 1991).

Ethnobotany

Bulrush is like the cattail in edibility, although it is purportedly sweeter. Young shoots coming up in the spring can be eaten raw or cooked. Bulrush pollen is eaten as flour in bread, mush, or pancakes. Later in the season, the seeds can be beaten in baskets or pails, ground into a meal and used as flour. The large rhizomes are eaten raw or cooked; sometimes they are dried in the sun, then pounded into a kind of flour. Softstem bulrush (*Schoenoplectus tabernaemontani*), a similar species, has as much as 8% sugar and 5.5% starch in rhizomes, but less than 1% protein (Harrington, 1972).



California bulrush growing in a Louisiana wetland.
Photo by: Garret Thomassie, USDA-NRCS Golden Meadow Plant Material Center, Galliano, Louisiana.

The rhizome (underground stem) is used for the black element in basket design. Rhizomes are obtained by digging around the plant and following them out from the parent plant. Often the green stalks are cut, to make the rooting area more accessible. Bulrushes are called black root by Pomo basket weavers in California; the cream-colored rhizome is dyed black for basketry designs (Hickman, 1993).

Tule houses were common throughout many parts of California with the overlapping tule matters being well-insulated and rain-proof. Willow poles, arched and anchored into the ground and tied with cordage or bark, formed the framework. The walls were thatched with mats of tule or cattail and secured to the frame. In Nevada, tules and willows were bound together in a sort of crude weaving for "Kani", the Paiute name for summerhouse (Murphy, 1959).

Tules and cattails were used as insulating thatch for structures, matting, bedding, and roofing materials. Shredded tule was used for baby diapers, bedding, and menstrual padding. Women made skirts from tule. During inclement weather, men wore shredded tule capes, which tied around the neck and belted at the waist. Duck decoys were made of tule. Several California Indian tribes made canoes of tule stems bound together with vines from wild grape. Groups located near the California coast used tule to make large round mud-shoes so they could walk without sinking (Murphy, 1959).

Wildlife: The seeds, being less hairy and larger than cattail, are one of the most important and commonly used foods for ducks and for certain marsh birds and shorebirds (Martin et al., 1951). Bulrushes provide choice food for wetland birds: American widgeon, bufflehead, mallard, pintail, shoveler, blue-winged teal, cinnamon teal, greater scaup, lesser scaup, avocet, marbled godwit, clapper rail, Virginia rail, sora rail, long-billed dowitcher, and tricolored blackbird. Canada geese and white-fronted geese prefer the shoots and roots. The stems provide nesting habitat for blackbirds and marsh wrens (Martin et al., 1951).

Freshwater emergent wetlands are among the most productive wildlife habitats in California. They provide food, cover, and water for more than 160 species of birds and numerous mammals, reptiles and amphibians. The endangered Santa Cruz long-toed salamander and rare giant garter snake use these wetlands as primary habitat. (Hickman, 1993).

Status

Threatened or Endangered: California bulrush is not listed as threatened or endangered (US Fish and Wildlife Service, 2025).

Wetland Indicator: California bulrush is an obligate wetland indicator in the Arid West, Atlantic and Gulf Coastal Plain, Eastern Mountains and Piedmont, Great Plains, Hawaii, Midwest and Western Mountains, Valleys, and Coast.

Weedy or Invasive: Currently, no evidence of invasiveness has been exhibited.

Please consult the PLANTS Web site and your State Department of Natural Resources for this plant's status, such as, state noxious status and wetland indicator values.

Planting Guidelines

As a rule, California bulrush can be planted between April 1 and September 30. Other considerations include:

- California bulrush can be planted anytime past the last frost date if there is a need to plant earlier and available transplants are actively growing. In some areas this may be earlier than April 1.
- Newly established transplants are more susceptible to salt damage. Planting dates should be timed to avoid seasonally high levels of salt.
- In interior marshes with poor water circulation, avoid planting between mid-July and the end of August. Elevated water temperatures are generally detrimental to new transplants, limit July and August plantings to lakes, bayous, and other areas of frequent tidal exchange.
- Late fall plantings in October and November have been successful, but they should be limited to sites that are well protected and have minimal winter storm effect.

Management

California bulrush will thrive in a wide variety of marsh sites from mud flats to open water. Hydrology is the most important factor in determining wetland type, revegetation success, and wetland function and value. Changes in water levels influence species composition, structure, and distribution of plant communities. Water management is critical during plant establishment and remains crucial throughout the life of the wetland for proper community management. Heavy grazing will eliminate *Schoenoplectus* species as well as other native species (Martin et al., 1951).

Muskrats have evolved with wetland ecosystems and form a valuable component of healthy functioning wetland communities. Muskrats use emergent wetland vegetation for hut construction and for food. An area of open water is created

around the huts, which increase wetland diversity by providing opportunities for aquatic vegetation to become established. Muskrats opening the dense stands of emergent vegetation also create habitat for other species. Both beaver and muskrats often improve wetland habitat (Dahl & Johnson, 1991).

Traditional Resource Management: The plant must grow in coarse-textured soil that is free of gravel, clay, and silt for the roots to be of the quality necessary for basketweaving. Plants are tended by gathering rhizomes and reducing plant density. Sustainable harvesting of plants occurs through limiting harvest in any given area. Fire was also used to manage tule wetlands. Fire eliminated decadent old stems and restored open water to the wetland. This stimulates growth of new shoots from rhizomes and provides a bare soil substrate for seed germination. Many Native Americans feel the use of herbicides is inappropriate in traditional gathering sites. Bulrush is densely rhizomatous with abundant seed production. In most cases, it will out-compete other species within the wetland area of the site, eliminating the need for manual or chemical control of invasive species (Timbrook, 1997; Murphy, 1959; Roos-Collins, 1990; Clarke, 1977).

Pests and Potential Problem

Ducks and other birds consume seeds. Geese, muskrats and nutria consume the rhizomes and early shoots. Nutria is the greatest cause of problems in California bulrush establishment, as they excavate plants to feed on the roots. Where nutria is a problem, well-anchored wire cages are required to allow plants to become established (Tiner, 1984).

Environmental Concerns

California bulrush forms dense patches and can out compete other native plants for resources. It can displace native plants resulting in a reduction in plant diversity. Effects on agriculture have been seen when water flow is impeded in irrigation channels due to dense growth. Toxicity is a concern; however, no toxicity has been shown to date (Dahl & Johnson, 1991).

Control

Control methods include manual removal, mulching, solarization, and biological controls when available. Chemical control via herbicides with spot treatment and foliar application post-emergence have been used (Hickman, 1993). Please contact your local agricultural extension specialist or county weed specialist to learn what works best in your area and how to use it safely. Always read labels and safety instructions for each control method.

Seeds and Plant Production

Seed Collections: Select seed collection sites where continuous stands with few intermixed species are found. At each collection location, obtain permission for seed collection. Seed is harvested by either taking hand clippers or cutting the stem off below the seed heads or stripping the seed heads off the stalk. Less than ½ hour is required to make a decent collection of 1 to 2 cups of seed. The ease of collection is affected by water depth. Collect and store seeds in brown paper bags or burlap bags. Dry the seeds in these bags. Seeds and seed heads need to be cleaned in a seed cleaner. Plant cleaned seed in fall into a clean, weed-free, moist seed bed. Flooded or ponded soil will significantly increase seedling mortality. Broadcast seed and roll in or rake ¼ " to ½ " from the soil surface. Some seeds may be lost due to scour or flooding. California bulrush seed quantity has been reported as 36,000,000 per pound (USDA NRCS, 2003).

Seed germination in greenhouse: Clean seed - blow out light seed with an air screen cleaner or other mechanism. To germinate seeds, plant in greenhouse in 1" x 1" x 2" pots, ¼ " under the soil surface. Keep the soil surface moist at a temperature of 100 degrees F (plus or minus 5 degrees). Seeds begin to germinate after a few weeks in warm temperatures. Plant plugs are ready in 100-120 days. Plant seeds in August for plugs ready for planting in November. These plants are very small; growing plants to a larger size will result in increased re-vegetation success (Hoag & Sellers, 1995).

Cultivars, Improved, and Selected Materials (and area of origin)

This species is available from most nurseries that offer wetland plants.

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